

CS 201 - Data Structures II

Introduction

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Outline

- Data Structure – What, Why?
- Important Considerations
- Course Overview
 - Learning Outcomes
 - Weekly Content
 - Assessments
- Course Logistics

What is data Structure?

- A **data structure** is a way of organizing and storing data to perform operations efficiently.

Organization in real-world



Some applications

- Consider the applications in:
 - Web Search
 - Database queries
 - Machine Learning
 - Computer Graphics
 - Routing/Navigation
 -
 - and many more

Commonly used data structures

- Array
- Linked List
- Stack
- Queue
- Tree
- Graph
- Hash table

Three things to consider for a data structure

Correctness: The data structure should correctly implement its interface.

Time complexity: The running times of operations on the data structure should be as small as possible.

Space complexity: The data structure should use as little memory as possible.

Need for efficiency

- Consider a linear implementation of data (i.e. a list). Suppose that we have a list of one million (10^6) items.
- Consider an application which has to access each of these 10^6 items at least once.
- If we have to scan the entire list to find each data item, then the total number of accesses will be $10^6 \times 10^6 = 10^{12}$
- With today's computers performing $\sim 10^9$ operations per second, it will take approximately $10^{12} / 10^9 = 10^3 = 1000$ seconds = 16 min 40 seconds to complete the entire operation.
- Clearly, there is a need for an efficient data structure for this application.

Look at the other side!

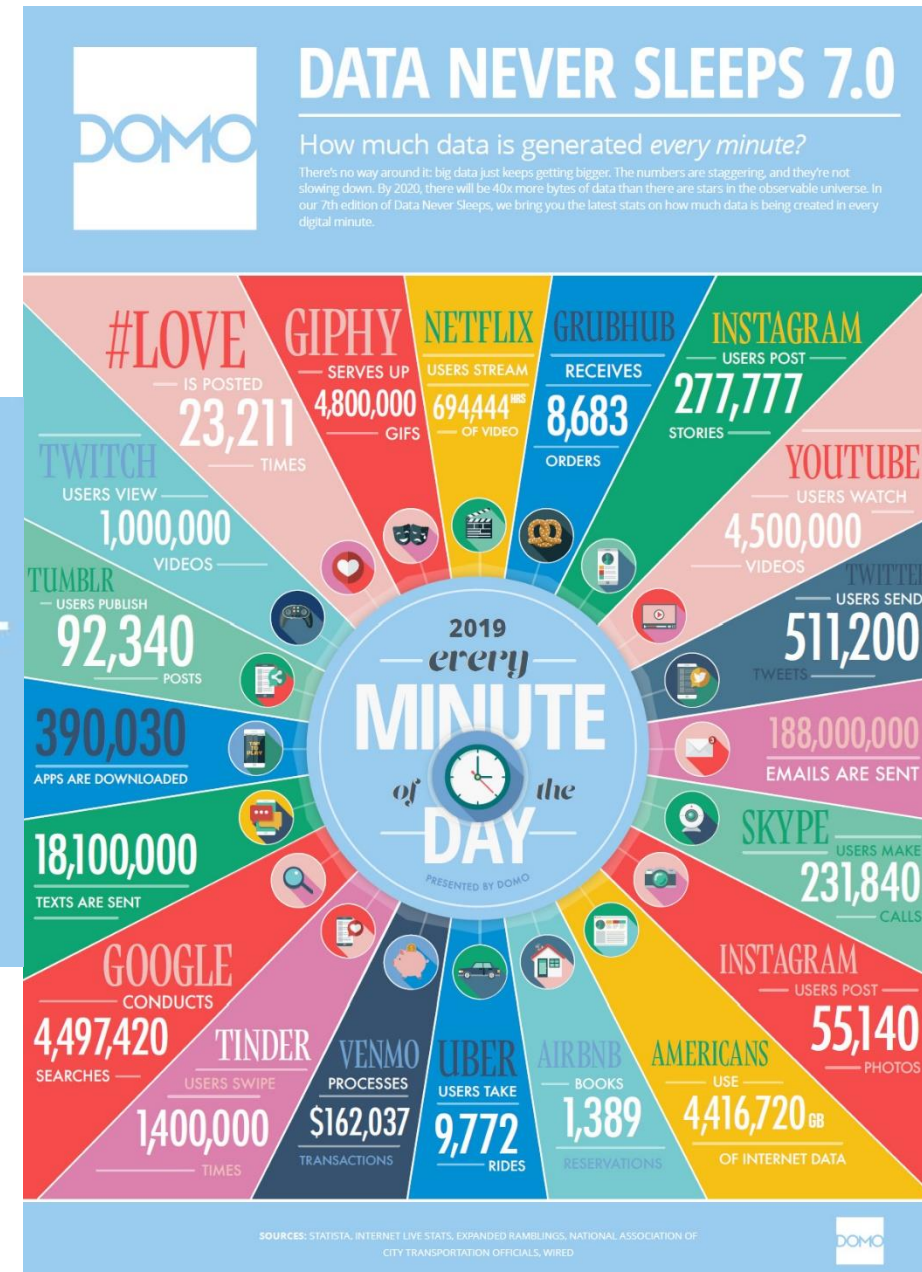
DATA NEVER SLEEPS 7.0

How much data is generated *every minute*?

There's no way around it: big data just keeps getting bigger. The numbers are staggering, and they're not slowing down. By 2020, there will be 40x more bytes of data than there are stars in the observable universe. In our 7th edition of Data Never Sleeps, we bring you the latest stats on how much data is being created in every digital minute.

This course is not about BIG data!

[Infographic: Why Big Data Keeps Getting Bigger \(visualcapitalist.com\)](https://visualcapitalist.com)



Knowing a data structure

- Knowing a data structures means:
 - understand the underlying structure
 - key operations (retrieval, traversal, add, remove etc.) and their complexity
 - ability to implement the data structure
 - strength and weaknesses
- Given a situation, you should be able to decide on the best data structure that fulfils the need

Course Syllabus

Thanks